

DSC550Week6.2.1

July 13, 2025

0.1 1. Data Loading, Cleaning, and Preparation

```
[22]: import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score, mean_squared_error
from sklearn.preprocessing import StandardScaler, QuantileTransformer
from sklearn.decomposition import PCA
import numpy as np

# Load the data
try:
    df = pd.read_csv('./house-prices-advanced-regression-techniques/train.csv')
except FileNotFoundError:
    print("Error: housing_data.csv not found. Make sure the file is in the same_
    ↪directory or provide the correct path.")
    exit()

# Drop the "Id" column
df = df.drop("Id", axis=1)

# Handle missing values
for col in df.columns:
    if df[col].isnull().sum() / len(df) > 0.4:
        df = df.drop(col, axis=1)

for col in df.select_dtypes(include=np.number).columns:
    df[col] = df[col].fillna(df[col].median())

for col in df.select_dtypes(exclude=np.number).columns:
    df[col] = df[col].fillna(df[col].mode()[0])

# Convert categorical features to dummy variables
df = pd.get_dummies(df, drop_first=True)

# Split into training and testing sets
X = df.drop("SalePrice", axis=1)
y = df["SalePrice"]
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
↳random_state=42)
```

0.2 2. Linear Regression (Original Data)

```
[25]: # Linear Regression
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
r2 = r2_score(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))

print(f"Linear Regression (Original Data):\nR2: {r2:.4f}\nRMSE: {rmse:.4f}")
```

Linear Regression (Original Data):
R2: 0.6478
RMSE: 51973.1381

0.3 3. PCA Transformation

```
[28]: # PCA
pca = PCA(n_components=0.9, random_state=42)
pca.fit(X_train)
X_train_pca = pca.transform(X_train)
X_test_pca = pca.transform(X_test)
num_features_pca = X_train_pca.shape[1]
print(f"Number of features after PCA: {num_features_pca}")
```

Number of features after PCA: 1

0.4 4. Linear Regression (PCA Transformed Data)

```
[31]: # Linear Regression (PCA)
model_pca = LinearRegression()
model_pca.fit(X_train_pca, y_train)
y_pred_pca = model_pca.predict(X_test_pca)
r2_pca = r2_score(y_test, y_pred_pca)
rmse_pca = np.sqrt(mean_squared_error(y_test, y_pred_pca))

print(f"Linear Regression (PCA):\nR2: {r2_pca:.4f}\nRMSE: {rmse_pca:.4f}")
```

Linear Regression (PCA):
R2: 0.0635
RMSE: 84754.5802

0.5 5. Min-Max Scaling and Feature Selection

```
[34]: # Min-Max Scaling
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Variance thresholding
variances = np.var(X_train_scaled, axis=0)
high_variance_indices = variances > 0.1
X_train_high_variance = X_train_scaled[:, high_variance_indices]
X_test_high_variance = X_test_scaled[:, high_variance_indices]

print(f"Number of features with variance > 0.1: {X_train_high_variance.
      ↪shape[1]}")
```

Number of features with variance > 0.1: 227

```
[ ]:
```